# Semantic-Guided Fusion Network for Multi-source Remote Sensing Image Classification

Yuwei Zhao, Chuanzheng Gong, Baogui Huan, Feng Gao, *Member, IEEE*, Junyu Dong, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*—Multi-source remote sensing image classification has attracted increasing attention due to the complementary spectral, structural, and geometric information. However, existing methods still suffer from two limitations: insufficient semantic contextual modeling and unreliable feature fusion caused by slight spatial misalignment. To address these issues, we propose a Semantic-Guided Fusion Network (SGFNet) for multi-source remote sensing image classification. Specifically, the *Semantic Mixing Convolution Block (SMCB)* is designed to dynamically generate semantic-aware convolution kernels according to contextual relationships among feature representations, thereby enhancing semantic dependency modeling and adaptive feature aggregation. In addition, the *Frequency Modulated Fusion Block (FMFB)* is introduced to perform cross-modal interaction in the frequency domain, which effectively alleviates the influence of slight spatial misalignment and improves complementary information fusion. Extensive experiments conducted on the Augsburg and Houston 2018 datasets demonstrate that the proposed SGFNet consistently outperforms several state-of-the-art methods. The code will be made publicly available at https://github.com/oucailab/SGFNet.**

***Index Terms*—Multi-source data classification; Hyperspectral image; Light detection and ranging; Synthetic aperture radar; Multi-source data fusion.**

## I. Introduction

WITH the continuous growth of satellite missions and sensor technologies, massive volumes of high-resolution and multi-source remote sensing data are now available, providing unprecedented opportunities to monitor the Earth's surface in a timely and comprehensive manner. Accurately identifying and mapping land-cover types is essential for urban planning, precision agriculture, and disaster management [1].

With the rapid development of Earth observation technologies, a large volume of multi-source remote sensing data has become increasingly accessible. Hyperspectral images (HSIs) capture detailed spectral information across hundreds of contiguous wavelength bands. This rich spectral resolution makes HSI particularly valuable for land cover classification. However, HSI is susceptible to noise and atmospheric interference, which reduces data quality and reliability. To mitigate these challenges, light detection and ranging (LiDAR) and synthetic aperture radar (SAR) data are frequently integrated as complementary sources [2]. LiDAR provides accurate, high-resolution 3D geometric and elevation information, while SAR offers valuable structural, scattering, and dielectric properties regardless of weather and lighting conditions. Consequently, the joint classification of HSI and LiDAR/SAR data has emerged as an effective paradigm to boost classification accuracy and robustness [3]. Therefore, in this letter, we mainly focus on HSI and SAR/LiDAR joint classification.

Deep learning-based methods have gained considerable research interest for multi-source remote sensing image joint classification, due to their powerful capability in learning hierarchical feature representations and modeling complex nonlinear relationships [4]. By leveraging convolutional neural networks (CNNs) [5] [6], Transformers [7] [8], and more recently Mamba architectures [9], these methods can effectively extract discriminative spatial-spectral features from HSI while simultaneously capturing complementary structural and geometric information from SAR and LiDAR data.

Although existing methods have achieved remarkable classification performance, they still suffer from two limitations: ***1) Lacking adaptive semantic and contextual representation capability.*** Traditional convolutions use spatially shared and fixed kernels for all spatial locations. Such fixed kernels can hardly adapt to varying semantic structures and content-aware feature distributions. As a result, existing CNN-based methods often fail to effectively capture long-range semantic correlations in complex scenes. ***2) Slight cross-modal spatial misalignment weakens reliable feature fusion.*** Multi-source remote sensing data acquired by different sensors often suffer from slight spatial misalignment due to inconsistent imaging geometries and sensor characteristics. Although the positional deviations are usually small, local regions and object boundaries may still exhibit structural inconsistencies across modalities. Such misalignment makes direct spatial feature fusion unreliable.

To effectively overcome these limitations, we propose the **S**emantic-**G**uided **F**usion **Net**work (**SGFNet**) for multi-source data joint classification. Specifically, to enhance the semantic and contextual feature representations for HSI and SAR/LiDAR data, we propose the *Semantic Mixing Convolution Block (SMCB)*. It dynamically generates semantic-aware convolution kernels according to contextual relationships among feature representations. By enabling position-specific contextual modeling and selectively emphasizing semantically relevant information, the proposed module effectively enhances global semantic dependencies. In addition, we introduce the *Frequency Modulated Fusion Block (FMFB)*

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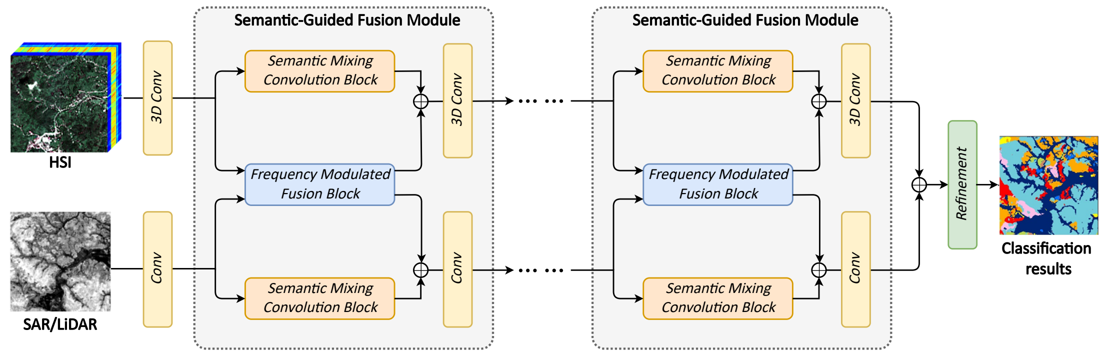


Fig. 1. Overview of the Semantic-Guided Fusion Network (SGFNet). It adopts a dual-branch hierarchical architecture composed of multiple stacked Semantic-Guided Fusion Modules (SGFMs). Within each SGFM, two Semantic Mixing Convolution Blocks (SMCB) are introduced to model semantic-aware contextual representations. These blocks enhance feature interaction by dynamically integrating global semantic context into local convolution. To further strengthen cross-modal interaction, a Frequency Modulated Fusion Block (FMFB) is inserted between the two branches. It performs frequency-aware feature modulation and adaptive information exchange across modalities.

to alleviate the slight spatial misalignment between multi-source data. It performs cross-modal feature interaction in the frequency domain rather than directly conducting spatial-domain fusion. Through adaptive frequency-aware modulation, our FMFB effectively enhances complementary information integration and improves the discriminative capability of multi-source feature representations.

Our main contributions are summarized as follows:

- **Feature Enhancement.** We propose SMCB to enhance semantic and contextual feature representations. Unlike conventional convolutions with spatially shared fixed kernels, our SMCB uses semantic-aware dynamic convolution to improve contextual dependency modeling.
- **Cross-Modal Interaction.** We introduce FMFB to alleviate the slight spatial misalignment. By performing adaptive cross-modal interaction in the frequency domain, our FMFB enhances complementary information integration.
- **Experimental Contribution.** Extensive experiments conducted on two benchmark datasets demonstrate that the proposed SGFNet consistently outperforms several state-of-the-art methods. In addition, the source codes will be publicly released to facilitate future research on multi-source data fusion.

## II. Methodology

### A. Overall Framework of the Proposed SGFNet

The overall framework of the proposed Semantic-Guided Fusion Network (SGFNet) is illustrated in Fig. 1. The framework adopts a dual-branch architecture, where the HSI and SAR/LiDAR data are first processed separately through convolutional layers to extract preliminary spectral–spatial and structural representations. To achieve effective cross-modal interaction, the network introduces multiple stacked Semantic-Guided Fusion Modules (SGFM). Within each module, a Semantic Mixing Convolution Block (SMCB) is employed to capture high-level semantic contextual information and enhance discriminative feature learning. Meanwhile, a Frequency Modulated Fusion Block (FMFB) is designed to model complementary frequency-domain characteristics between heterogeneous modalities, enabling the network to preserve fine-grained structural details and suppress redundant or noisy information. The semantic and frequency-aware features are then adaptively aggregated through residual fusion operations, promoting more robust cross-modal feature interaction.

By progressively stacking several SGFMs, the framework performs hierarchical semantic refinement and deep multi-modal feature fusion. Finally, the fused representations are integrated through two convolutional layers for refinement to further enhance category consistency and discrimination, producing the final classification results.

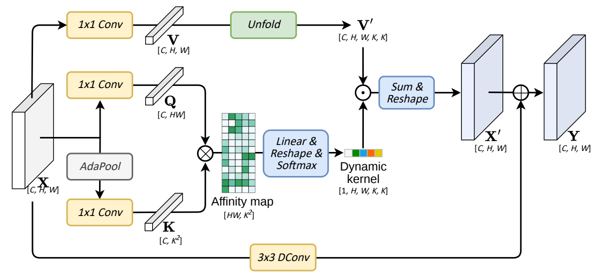


Fig. 2. Illustration of the Semantic Mixing Convolution Block (SMCB).

### B. Semantic Mixing Convolution Block (SMCB)

Standard convolutions use spatially shared fixed kernels, limiting their ability to adapt to varying semantic structures and contextual relationships. To address this issue, we propose the Semantic Mixing Convolution Block (SMCB), which integrates convolution with semantic-aware dynamic filtering. Instead of fixed kernels, SMCB dynamically generates spatially adaptive kernels based on semantic affinities between query and key features. The resulting dynamic kernels guide local feature aggregation, enabling the network to emphasize semantically relevant contextual information adaptively. Details of the SMCB are shown in Fig. 2. Given an input feature $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, it is projected by a $1\times1$ convolution to obtain the value and query features as:

$$\mathbf{V} = f_v(\mathbf{X}), \quad \mathbf{Q} = f_q(\mathbf{X}), \tag{1}$$

where $f_v$ and $f_q$ denote the convolution layer. Meanwhile, $\mathbf{X}$ is passed through an adaptive pooling layer to extract compact contextual information as:

$$\mathbf{K} = f_k(\text{AdaPool}(\mathbf{X})), \tag{2}$$

where $f_k$ denotes the convolution layer. We obtain $\mathbf{Q} \in \mathbb{R}^{C \times HW}$ and $\mathbf{K} \in \mathbb{R}^{C \times K^2}$ via reshaping. Here $K$ denotes the dynamic kernel size. The query and key features are multiplied to obtain an affinity map:

$$\mathbf{A} = \mathbf{Q}^T \mathbf{K}. \tag{3}$$

This affinity map measures the relationship between each spatial position and the $K^2$ sampling locations in a local window. It is then processed by a linear layer, reshaped, and normalized by Softmax to generate the dynamic convolution kernel:

$$\mathbf{D} = \text{Softmax}(\text{Reshape}(\text{Linear}(\mathbf{A}))). \tag{4}$$

Specifically, given the input feature $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, the query is reshaped to $\mathbf{Q} \in \mathbb{R}^{C \times HW}$, and the key is obtained by adaptively pooling $\mathbf{X}$ to a $K \times K$ spatial resolution followed by a $1 \times 1$ convolution, then flattened to $\mathbf{K} \in \mathbb{R}^{C \times K^2}$. The resulting affinity map $\mathbf{A} = \mathbf{Q}^\top \mathbf{K} \in \mathbb{R}^{HW \times K^2}$ measures the relationship between each of the $HW$ spatial positions and the $K^2$ globally-pooled semantic anchors. This affinity map is then processed by a linear layer that preserves the $K^2$ dimensionality while allowing learnable re-weighting, reshaped into $\mathbb{R}^{1 \times H \times W \times K \times K}$, and normalized via Softmax along the last two dimensions to ensure the sampling weights at each spatial position sum to one, yielding the final dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$. Note that $\mathbf{D}$ is shared across all $C$ channels and is broadcast when applied to the unfolded value feature $\mathbf{V}' \in \mathbb{R}^{C \times H \times W \times K \times K}$ in Eq. (6). {R1-C4}

In parallel, the value feature $\mathbf{V}$ is unfolded into local patches:

$$\mathbf{V}' = \text{Unfold}(\mathbf{V}). \tag{5}$$

The dynamic kernel $\mathbf{D}$ is then applied to the unfolded value feature. In each channel of $\mathbf{V}'$, for every spatial position $(h, w)$, the output feature is computed by weighted summation over its local $K \times K$ neighborhood as.

$$\mathbf{X}' = \sum_{i=1}^{K} \sum_{j=1}^{K} \mathbf{D}_{h,w,i,j} \cdot \mathbf{V}'_{h,w,i,j}. \tag{6}$$

Besides the dynamic context branch, the framework also includes a local feature enhancement branch. The original input $\mathbf{X}$ is processed by a $3 \times 3$ depth-wise convolution to compute $\mathbf{X}_d$. Finally, the dynamically aggregated feature $\mathbf{X}'$ is combined with the local depth-wise convolution feature through residual addition to produce the final output:

$$\mathbf{Y} = \mathbf{X}' + \mathbf{X}_d. \tag{7}$$

Compared with prior dynamic convolution methods (e.g., CondConv [10], Involution [11]), which generate content-adaptive kernels purely from the local receptive field of each spatial position, SMCB instead computes the dynamic kernel by measuring the affinity between a per-position query $\mathbf{Q}$ and a set of globally pooled key vectors $\mathbf{K}$ (Eq. 2). This allows each spatial location to adaptively aggregate information conditioned on a compact, globally-informed semantic summary rather than being restricted to a fixed local window, which is the primary source of SMCB's long-range dependency modeling capability. Furthermore, unlike existing semantic-aware filtering approaches that rely on externally supervised or heuristically defined semantic categories to modulate the convolution kernels, SMCB requires no auxiliary semantic supervision: the semantic affinity in Eq. (3) is learned entirely end-to-end from feature-level query-key interactions, making it directly applicable to multi-source remote sensing data where reliable semantic labels for intermediate features are unavailable. In addition, rather than generating a full $C \times K \times K$ kernel per spatial position as in conventional dynamic filtering, SMCB decouples kernel generation from the channel dimension: the dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$ is shared across channels and only modulates spatial sampling weights, while channel-wise transformation is handled separately by the $1 \times 1$ projections in Eq. (1)–(2). This design substantially reduces the parameter and computational overhead relative to fully dynamic per-channel kernels, as quantified in Section III-D. {R1-C1} Physically, the key vectors $\mathbf{K}$ in Eq. (2) are obtained by adaptively pooling contextual information from across the entire spatial extent of the input feature map, and therefore represent global contextual descriptors rather than locally restricted ones. When computing the affinity map in Eq. (3), every spatial position measures its relevance to these globally-derived key vectors regardless of their physical distance, allowing the resulting dynamic kernel to adaptively aggregate information from semantically related regions located anywhere in the image. This effectively extends the receptive field of the dynamic kernel from a local spatial window to the semantic structure of the entire feature map, which constitutes the underlying physical mechanism through which SMCB models long-range semantic dependencies. {R2-C3}

The SMCB achieves a more flexible and semantically aware feature extraction mechanism. It is particularly suitable for understanding the complex structures of remote sensing data, where both local structural details and semantic contextual dependencies are critical.

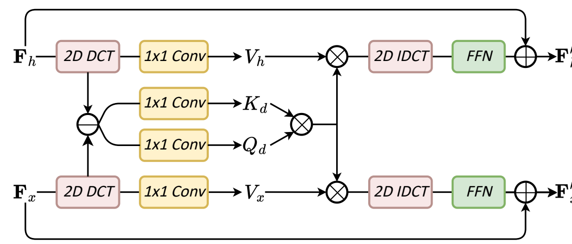


Fig. 3. Illustration of the Frequency Modulated Fusion Block (FMFB).

## C. Frequency Modulated Fusion Block

Multi-source data are usually collected by different sensors with varying imaging geometries and acquisition times, which often results in slight spatial misalignment between modalities. Local object regions may exhibit small positional deviations and structural inconsistencies. Such slight misalignment may weaken the effectiveness of cross-modal feature fusion. To this end, we propose the Frequency Modulated Fusion Block (FMFB), which performs adaptive cross-modal fusion in the frequency domain to enhance feature consistency.

Details of FMFB are illustrated in Fig. 3. It takes the feature representations $\mathbf{F}_h$ from the HSI branch and $\mathbf{F}_x$ from the SAR/LiDAR branch as inputs. It first applies 2D Discrete Cosine Transform (DCT) to project both features into the frequency domain:

$$\hat{\mathbf{F}}_h = \mathrm{DCT}(\mathbf{F}_h), \quad \hat{\mathbf{F}}_x = \mathrm{DCT}(\mathbf{F}_x). \tag{8}$$

We then compute the element-wise difference in the frequency domain:

$$\hat{\mathbf{F}}_d = \hat{\mathbf{F}}_h - \hat{\mathbf{F}}_x. \tag{9}$$

Based on this joint representation, two $1 \times 1$ convolution layers generate the query and key features as $\mathbf{Q}_d = f_q(\hat{\mathbf{F}}_d)$, $\mathbf{K}_d = f_k(\hat{\mathbf{F}}_d)$.

Meanwhile, the two DCT-transformed inputs are separately projected into value features:

$$\mathbf{V}_h = f_v^h(\hat{\mathbf{F}}_h), \quad \mathbf{V}_x = f_v^x(\hat{\mathbf{F}}_x). \tag{10}$$

We emphasize that the frequency-domain difference feature $\hat{\mathbf{F}}_d$ is used exclusively to generate the query and key features that guide the cross-modal attention weights in Eqs. (11)–(12), while the value features $\mathbf{V}_h$ and $\mathbf{V}_x$, which carry the actual modality-specific content, are separately projected directly from the original DCT-transformed features $\hat{\mathbf{F}}_h$ and $\hat{\mathbf{F}}_x$ without passing through the difference operation. As such, $\hat{\mathbf{F}}_d$ only modulates how attention is allocated across frequency components based on cross-modal discrepancy, rather than transforming the content that is ultimately propagated and fused. This design ensures that no modality-specific information is discarded or suppressed by the difference operation. {R3-C6}

Next, attention is computed as follows:

$$\mathbf{F}_h^{attn} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_d^\top}{\sqrt{d_k}}\right)\mathbf{V}_h, \tag{11}$$

$$\mathbf{F}_x^{attn} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_d^\top}{\sqrt{d_k}}\right)\mathbf{V}_x, \tag{12}$$

where $d_k$ denotes the channel dimension of $\mathbf{Q}_d$ and $\mathbf{K}_d$, used as a scaling factor to stabilize gradients during training. {R1-C3}

After that, the modulated frequency-domain features are transformed back into the spatial domain through 2D inverse DCT:

$$\hat{\mathbf{F}}'_h = \mathrm{IDCT}(\mathbf{F}_h^{attn}), \quad \hat{\mathbf{F}}'_x = \mathrm{IDCT}(\mathbf{F}_x^{attn}). \tag{13}$$

The reconstructed feature is further refined by the Feed-Forward Network (FFN) and residual connections:

$$\mathbf{F}'_h = \mathrm{FFN}(\hat{\mathbf{F}}'_h) + \mathbf{F}_h, \quad \mathbf{F}'_x = \mathrm{FFN}(\hat{\mathbf{F}}'_x) + \mathbf{F}_x. \tag{14}$$

TABLE I
EXPERIMENTAL RESULTS ON THE AUGSBURG DATASET. THE **BOLD** AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS. {R1-C5}

| Class | S$^2$ENet | DFINet | ExViT | DFFNet | MSFM | GCCQTNet | SGFNet |
|---|---|---|---|---|---|---|---|
| Forest | **98.10** | 97.38 | 90.04 | 96.17 | 97.17 | 95.26 | 97.18 |
| Residential area | **99.08** | 98.37 | 95.44 | 96.64 | 98.15 | 96.80 | 98.28 |
| Industrial area | 12.19 | 61.31 | 34.58 | 37.70 | 50.26 | **68.20** | 61.41 |
| Low plants | 91.78 | 92.63 | 90.68 | 94.21 | **95.52** | 91.35 | 95.48 |
| Allotment | 45.12 | 49.33 | 51.82 | 53.54 | 53.35 | 41.68 | **65.01** |
| Commercial area | 1.22 | 3.54 | **28.63** | 11.36 | 2.63 | 13.97 | 8.55 |
| Water | 24.09 | 26.61 | 17.65 | 27.15 | 49.97 | 46.80 | **56.62** |
| OA | 88.22 | 90.66 | 86.65 | 89.37 | 91.38 | 90.19 | **92.38** |
| AA | 53.08 | 61.31 | 58.41 | 59.54 | 63.31 | 64.87 | **68.93** |
| Kappa | 86.47 | 86.47 | 80.79 | 84.51 | 87.45 | 85.85 | **89.05** |

$$\mathbf{F}'_h = \mathrm{FFN}(\hat{\mathbf{F}}'_h) + \mathbf{F}_h, \quad \mathbf{F}'_x = \mathrm{FFN}(\hat{\mathbf{F}}'_x) + \mathbf{F}_x. \tag{15}$$

The rationale for performing cross-modal interaction in the frequency domain rather than the spatial domain lies in the shift theorem of the discrete transform: a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misalignment in multi-source remote sensing data typically manifests as small spatial translations or local geometric distortions, frequency-domain operations that primarily act on the energy distribution, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion. Moreover, the DCT concentrates most of the signal energy into low-frequency components, where the relative effect of small spatial misalignment is proportionally weaker than in high-frequency, edge-sensitive regions. This provides the theoretical basis for the improved robustness of FMFB to slight spatial misalignment between multi-source data. {R1-C2/R2-C2}

## III. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Datasets and Experimental Settings

The performance of the proposed SGFNet is evaluated on two benchmark multi-source remote sensing datasets for land-cover classification. The first dataset is the Augsburg Dataset [12], which consists of HSI and SAR data. It contains ground-truth annotations for seven land-cover classes and has a spatial size of 332 × 485 pixels. The second dataset is the Houston 2018 Dataset, which comprises HSI and LiDAR data. As it is obtained from the IEEE GRSS Data Fusion Contest 2018 [13], this dataset provides a representative multi-sensor optical geospatial benchmark for challenging urban land-cover and land-use classification tasks over the University of Houston campus and its surrounding areas. The dataset includes hyperspectral imagery with 48 spectral bands spanning wavelengths from 380-1050 nm, along with LiDAR measurements that provide accurate elevation information.

Our SGFNet was conducted on NVIDIA GeForce RTX 5080 GPU. The training phase spanned over 100 epochs. The AdamW optimizer was used with a learning rate of 0.001. The batch size was set to 32. To evaluate the performance of the proposed SGFNet, we compare it against other state-of-the-art methods, including S$^2$ENet [14], DFINet [15], ExViT

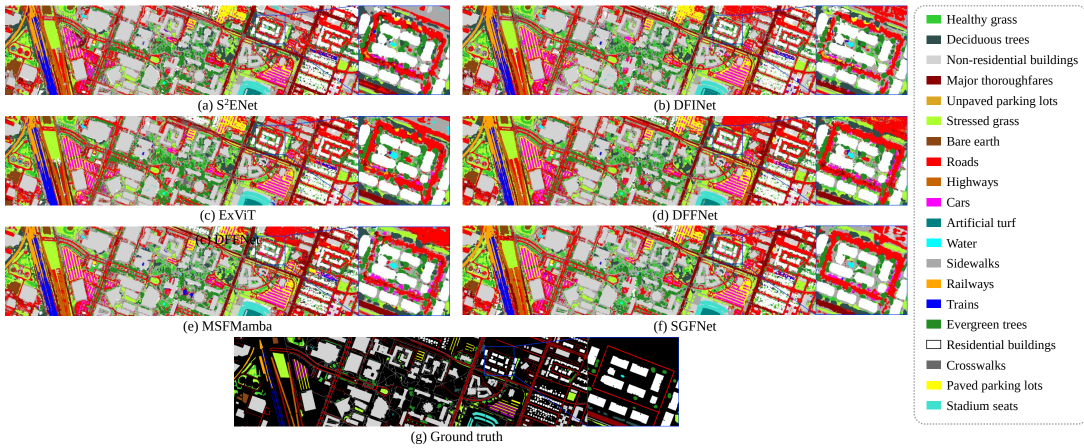


Fig. 4. Classification results of different methods on the Houston2018 dataset.

[16], DFFNet [17], and MSFM [18]. To quantitatively evaluate the experimental results, we employ three standard metrics: Overall Accuracy (OA), Average Accuracy (AA), and Kappa coefficient.

### B. Parameter Analysis

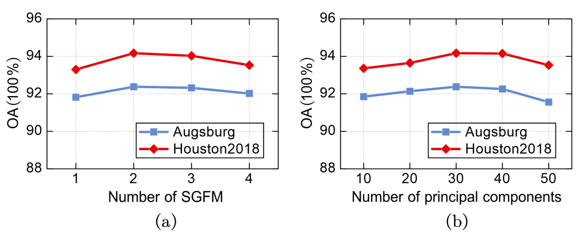


Fig. 5. Parameter analysis. (a) The relationship between OA and the number of SGFM. (b) The relationship between OA and the number of principal components after PCA for HSI.

**Number of SGFM.** The number of SGFM is an important parameter that may affect the classification performance. We test different numbers of SGFM from 1 to 4. The experimental results are shown in Fig. 5(a). When the number of SGFM is set to 2, our SGFNet achieves the best performance on both datasets. Therefore, in the following experiments, we use two SGFMs in the proposed SGFNet.

**Number of Principal Components for HSI.** HSI data contains rich spectral information but also introduces redundancy and spectral noise. Therefore, we use Principal Component Analysis (PCA) for HSI preprocessing. We tested the number of principal components for HSI, and the experimental results are shown in Fig. 5(b). It can be observed that when the number of principal components is set to 30, our SGFNet achieves the best classification performance. Retaining too few components discards crucial discriminative spectral features, leading to suboptimal accuracy. Thus, the number of principal components is empirically set to 30.

**Sensitivity of the Dynamic Kernel Size in SMCB.** The dynamic kernel size $K$ in SMCB determines the number of semantic sampling locations used to generate the dynamic convolution kernel, and may therefore affect the classification performance. We evaluate SGFNet with $K \in \{5, 7, 9, 11\}$ on both the Augsburg and Houston2018 datasets, and the results are shown in Table II. The OA on both datasets peaks at $K = 7$ (92.58% on Augsburg and 94.10% on Houston2018), and gradually declines as $K$ deviates from this value in either direction. A smaller $K$ provides insufficient semantic sampling locations for effective contextual modeling, while a larger $K$ introduces potentially redundant or noisy contextual information along with increased computational overhead. Based on this analysis, we set $K = 7$ in our experiments, as it achieves the best trade-off between classification accuracy and computational overhead. {R2-C4}

TABLE II
OA (%) OF SGFNET UNDER DIFFERENT DYNAMIC KERNEL SIZES $K$ IN SMCB.

| $K$ | 5 | 7 | 9 | 11 |
|---|---|---|---|---|
| Augsburg | 91.64 | **92.58** | 92.50 | 92.02 |
| Houston2018 | 93.26 | **94.10** | 94.03 | 93.47 |

We further analyze the sensitivity of SGFNet to the channel dimension $C$ in the Supplementary Material, which shows that SGFNet achieves the optimal balance between classification accuracy and model complexity when $C$ is set to 64.

### C. Experimental Results and Analysis

**Results on the Augsburg Dataset.** Table I shows the quantitative evaluations on the Augsburg dataset. The OA value of the proposed SGFNet achieves 92.38%, outperforming all the other methods. From the class-wise results, SGFNet achieves the best performance in Allotment, and remains highly competitive in the Industrial area. {R1-C5} Both categories usually exhibit complex spatial structures and significant spectral ambiguity, indicating that SGFNet can effectively capture discriminative semantic and contextual information. The corresponding classification maps on the Augsburg dataset are provided in the Supplementary Material due to the page limit of this Letter.

**Results on the Houston2018 Dataset.** The classification maps on the Houston2018 dataset are shown in Fig. 4, and

TABLE III
EXPERIMENTAL RESULTS ON THE HOUSTON2018 DATASET. THE **BOLD** AND UNDERLINE DENOTE THE BEST AND SECOND BEST RESULTS.
{R1-C5}

| Class | S$^2$ENet | DFINet | ExViT | DFFNet | MSFM | GCCQTNet | SGFNet |
|---|---|---|---|---|---|---|---|
| Healthy grass | **99.15** | 94.29 | 93.34 | 92.63 | 90.38 | 91.71 | 92.07 |
| Stressed grass | 84.35 | 92.71 | 93.44 | 92.44 | 94.04 | **94.22** | 93.18 |
| Artificial turf | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** |
| Evergreen trees | 98.13 | 98.92 | 95.87 | 99.38 | 99.34 | 98.58 | **99.76** |
| Deciduous trees | 86.39 | 97.77 | 96.85 | 99.16 | 99.11 | 98.71 | **99.94** |
| Bare earth | 99.33 | **100.0** | **100.0** | 99.96 | **100.0** | 99.96 | **100.0** |
| Water | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** |
| Residential buildings | 97.17 | 97.36 | 97.39 | 98.00 | 96.27 | 96.41 | **98.78** |
| Non-residential buildings | 93.93 | 93.29 | 93.00 | 93.73 | 95.58 | 94.16 | **96.24** |
| Roads | 70.17 | 75.73 | 72.18 | 80.31 | 75.90 | 77.79 | **85.33** |
| Sidewalks | 72.11 | 83.67 | 69.69 | 72.59 | 81.43 | 82.55 | **84.64** |
| Crosswalks | 82.47 | 94.23 | 88.12 | 91.84 | 96.22 | 97.11 | **98.14** |
| Major thoroughfares | **89.44** | 81.20 | 86.12 | 84.54 | 86.89 | 88.03 | 88.85 |
| Highways | 98.53 | 98.91 | **99.50** | 98.82 | 98.63 | 98.43 | 99.35 |
| Railways | 99.21 | **99.94** | 99.87 | 99.89 | 99.69 | 99.77 | 99.82 |
| Paved parking lots | 95.62 | 98.37 | 97.02 | 97.78 | 98.94 | **99.35** | 99.34 |
| Unpaved parking lots | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** | **100.0** |
| Cars | 96.77 | **99.09** | 98.13 | 98.89 | 97.81 | 97.44 | 98.67 |
| Trains | **100.0** | 99.46 | 99.98 | 99.91 | 99.97 | 99.89 | **100.0** |
| Stadium seats | **100.0** | 99.98 | 99.99 | **100.0** | **100.0** | 99.84 | **100.0** |
| OA | 90.05 | 91.02 | 89.98 | 91.21 | 92.38 | 92.13 | **94.17** |
| AA | 93.14 | 95.24 | 94.02 | 94.99 | 95.51 | 95.70 | **96.71** |
| Kappa | 87.20 | 88.46 | 87.14 | 88.68 | 90.16 | 89.86 | **92.46** |

the corresponding quantitative evaluations are illustrated in Table III. It can be observed that SGFNet produces classification maps that are more consistent with the ground truth, demonstrating superior land-cover discrimination capability. Table III shows that our SGFNet obtains the highest OA and Kappa, significantly outperforming other state-of-the-art methods. SGFNet achieves the best performance in several challenging categories, including Evergreen trees, Deciduous trees, Buildings, Roads, Sidewalks, Crosswalks, and Paved parking lots. These categories usually contain complex spatial distributions and strong spectral similarity with surrounding objects. The superior performance of SGFNet indicates that the proposed framework can effectively capture discriminative semantic information and exploit complementary features from multi-source data.

We note that the class imbalance observed in the above results primarily stems from the intrinsic, naturally imbalanced land-cover distribution of the benchmark datasets: classes such as Forest and Water occupy large, contiguous spatial extents and are represented by abundant labeled samples, whereas classes such as Commercial area and Industrial area correspond to sparse, scattered regions with comparatively few labeled samples. This inherent imbalance in sample availability, compounded by the spectral and structural similarity among man-made built-up surfaces, contributes to the under-performance on these minority classes discussed below. {R2-C1}

**Failure Case Analysis.** Although SGFNet achieves the best overall accuracy on both datasets, it does not consistently outperform all baselines on every individual class. On the Augsburg dataset, SGFNet underperforms the best method on the Industrial area and Commercial area classes, most notably on Commercial area (8.55% vs. 28.63% for ExViT). We attribute this to the severe class imbalance of these two categories, which limits the supervisory signal available for learning discriminative semantic patterns, as well as the strong spectral and structural similarity among man-made built-up surfaces (Commercial, Residential, and Industrial areas), which makes them intrinsically difficult to disambiguate. A preliminary exploration using focal loss and minority-class oversampling showed a modest improvement on these classes, suggesting that class imbalance is a contributing factor, though spectral ambiguity likely also plays a substantial role. On the Houston2018 dataset, SGFNet is marginally surpassed by GCCQTNet on the Paved parking lots class by a negligible margin (99.34% vs. 99.35%), and the relatively lower accuracy on Roads and Sidewalks reflects the weak discriminability of both HSI and SAR/LiDAR cues at the shared boundaries of paved urban surfaces. Nevertheless, SGFNet still achieves the highest AA among all compared methods on both datasets, indicating that it handles minority and spectrally ambiguous classes more effectively than all baselines on average, even though it does not dominate every single class. {R1-C6}

TABLE IV
ABLATION STUDY OF THE PROPOSED SGFNET.

| Model Variant | Augsburg | Houston2018 |
|---|---|---|
| Baseline | 88.54 | 89.65 |
| w/o SMCB | 89.88 | 91.01 |
| w/o FMFB | 91.29 | 92.37 |
| SGFNet (Full model) | **92.38** | **94.17** |

### D. Ablation Study

Table IV presents the ablation study results of the proposed SGFNet. The experiments are designed to evaluate the effectiveness of SMCB and FMFB. Starting from the baseline model, both modules consistently improve the classification performance. This demonstrates that SMCB can effectively enhance contextual representation capability, and FMFB effectively alleviates the slight spatial misalignment between multi-source data. The complete SGFNet achieves significant improvements of 3.84% and 4.52%, respectively. These results demonstrate that both proposed modules are complementary and jointly contribute to more robust multi-source feature fusion and classification. In addition, a detailed computational complexity analysis in the Supplementary Material shows that SGFNet achieves the smallest number of parameters and among the fastest inference speed of all compared methods, confirming a favorable trade-off between efficiency and accuracy. {R3-C1}

### E. Robustness to Spatial Misalignment

To empirically verify the robustness of FMFB to spatial misalignment, we conduct a controlled experiment on the Houston2018 dataset by artificially applying spatial shifts of $\{1, 2, 3, 5\}$ pixels along both the $x$- and $y$-directions to the LiDAR branch input, while keeping the HSI branch fixed. We directly evaluate the models trained on the original, well-aligned data under these shifted test conditions, and compare our SGFNet against two representative state-of-the-art methods, DFFNet and MSFMamba. As shown in Table V, the OA of DFFNet and MSFMamba degrades by 2.60% and 2.03%, respectively, as the shift increases from 0 to 5 pixels, whereas

SGFNet exhibits only a 1.40% drop, with the performance gap between SGFNet and the baselines widening progressively at larger shift magnitudes. This confirms that the proposed FMFB effectively alleviates the adverse impact of cross-modal spatial misalignment through frequency-domain modulation. {R1-C2/R2-C2}

TABLE V
COMPARISON OF OA (%) UNDER DIFFERENT DEGREES OF ARTIFICIAL SPATIAL SHIFTS ON THE HOUSTON2018 DATASET.

| Shift (pixels) | 0 | 1 | 2 | 3 | 5 |
|---|---|---|---|---|---|
| DFFNet | 91.21 | 90.98 | 90.09 | 89.36 | 88.61 |
| MSFMamba | 92.38 | 92.14 | 92.07 | 91.81 | 90.35 |
| **SGFNet (Ours)** | **94.17** | **94.03** | **93.96** | **93.64** | **92.77** |

## IV. CONCLUSIONS

In this letter, we propose SGFNet for multi-source remote sensing image classification. To address the limitation of traditional convolutions in modeling semantic contextual relationships, we propose Semantic Mixing Convolution Block (SMCB) to dynamically generate semantic-aware convolution kernels according to contextual affinities among feature representations. In addition, we introduce the Frequency Modulated Fusion Block (FMFB) to perform adaptive cross-modal interaction in the frequency domain. Extensive experiments on two benchmark datasets demonstrate that SGFNet consistently outperforms state-of-the-art methods. In future work, we plan to further investigate strategies for mitigating class imbalance, such as focal loss and minority-class oversampling, and study their interaction with the semantic-aware modeling in SMCB, in order to further improve the classification accuracy on minority and spectrally ambiguous classes. {R1-C6}

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